

Does Local Urban Governance Status Matter? Evidence from India

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Abstract

We exploit quasi-random variation around the multi-threshold criteria used to classify Census Towns (CTs) and focus on settlements near the thresholds that are likely to obtain statutory recognition. Using a local fuzzy regression discontinuity design and a multi-threshold criteria, we show that meeting the CT eligibility in 2001 raises the probability of statutory recognition by 2011. Instrumenting statutory recognition with CT eligibility, we estimate the effects of ULB status on local public goods provision: government schools increase by 13.86 (primary), 7.72 (middle), and 4.89 (secondary) units, healthcare infrastructure expands by 2.53 hospitals and 3.00 family welfare centers, and financial access deepens with 4.09 cooperative banks and 2.84 agricultural credit societies. Community amenities also improve, while sports infrastructure declines by 5.71 facilities, consistent with reallocation of urban land. The corresponding reduced-form estimates are directionally consistent and indicate that crossing the CT eligibility frontier improves public goods provision. Our findings indicate that timely municipalization of emerging urban areas can affect the provision of public goods in developing economies.

JEL Codes: H75, R53, O18

1 Introduction

Urban local bodies (ULBs) are responsible for a wide range of public goods in developing countries such as schools, primary health facilities, sanitation, streets, lighting, and local regulation. Accordingly, whether and how local urban governance status matters for local economic development is a long-standing question, with implications for many rapidly urbanizing economies ([Brollo et al., 2013](#); [Faguet, 2012](#); [Patrick and Mothorpe, 2017](#)). However, a key empirical challenge is that governance structures are endogenous: places that are bigger and more developed are more likely to be recognized as urban towns and receive municipal status, while smaller or less developed settlements may not receive such recognition and therefore stagnate even further by being outside the urban system ([Gadenne, 2017](#)). As a result, simple cross-sectional comparisons in such contexts will conflate pre-existing differences with the effects of policy. Put differently, governance status is chosen rather than randomly assigned, making causal inference difficult. To address this challenge, researchers have often relied on varying identification strategies. For instance, [Lee \(2021\)](#) finds a positive effect of urban governance on access to water facilities in India by developing a satellite-based measure of confounders such as urbanization and also controlling for population density and other demographic variables in an OLS framework with fixed effects. By contrast, [Mukhopadhyay \(2017\)](#) finds no significant effect of governance status on public goods, such as water and sanitation, in India by comparing the provision of public goods across different types of urban settlements as categorized by their governance status. Meanwhile, although most papers study the impact on the extensive margin, more recently, [Narasimhan and Weaver \(2024\)](#) focus on the intensive margin of public goods provision, examining how governance status may have differential effects on public goods provision depending on the size of the settlement. More broadly, other studies have also explored this question in different contexts and using various methods ([Mukhopadhyay, 2015](#); [Mitra and Nagar, 2018](#)), yet the empirical evidence remains mixed.

This challenge is salient in India, where a large number of settlements are urban in

enumeration long before they are urban in *governance*: they meet the Census definition of an urban area yet continue to be administered as villages. We refer to this wedge between functional urbanization and formal urban governance as the **urban governance gap**. The gap is large and policy-relevant: of the roughly 5,765 settlements that met all three Census Town criteria in 2001, only about 44% had been granted statutory urban status by 2011. The issue has drawn the attention of policymakers. For instance, the Ministry of Housing and Urban Affairs has urged states to convert eligible Census Towns into statutory towns ([Ministry of Housing and Urban Affairs, Government of India, 2016](#)) and a recent World Bank report prepared for India’s 16th Central Finance Commission explicitly calls for incentivizing this transition ([World Bank, 2024](#)). Whether closing this gap actually improves local public goods provision remains an open question, and the lack of credible empirical estimates of the effect of local urban governance status on public goods provision has limited the ability of policymakers to design effective strategies to address this central issue.

This paper leverages a sharp institutional feature in India to address this identification problem: the Census Town (CT) classification thresholds that determine eligibility for an area to be considered urban. The Indian Census defines Census Towns as settlements meeting all three criteria: (i) population of at least 5,000 (ii) population density of at least 400 per square kilometer and (iii) at least 75% of the male main workforce in non-agricultural activities. Crossing these multi-dimensional cutoffs sharply increases the probability that a settlement will be formally recognized by state governments as a Statutory Town (ST), i.e. granted a municipality or other ULB status, but notably, it *does not* guarantee it. We exploit this quasi-random variation in recognition likelihood around the thresholds to construct a local fuzzy regression discontinuity (RD) design. In effect, meeting the Census Town criteria in 2001 serves as an instrument for obtaining statutory recognition in 2011, allowing us to isolate the effect of local urban governance status on public goods offered by the settlements.

The core idea is a fuzzy multi-threshold RD with multiple running variables. Using rich settlement-level microdata from the Socioeconomic High-resolution Rural-Urban Geographic

(SHRUG) platform (Asher, Novosad et al., 2019), we create a running variable called frontier distance using the three CT cutoffs and restrict attention to settlements near these thresholds. We first confirm a strong first-stage relationship: becoming marginally eligible as a Census Town in 2001 leads to a jump in the probability of statutory recognition in 2011. We then estimate the effect of statutory recognition, i.e., ULB status, on a broad set of development indicators measured by the 2011 Indian Census. Our outcome data are assembled by harmonizing the 2011 Village Directory (VD) and Town Directory (TD), which enumerate local infrastructure and amenities. We link these to demographic data from the 2011 Primary Census Abstract (PCA). This provides a comprehensive settlement-level panel of outcomes including educational facilities, health infrastructure, financial institutions, and other community amenities.

Preview of results. We find that crossing the urban eligibility thresholds leads to a sharp increase in the likelihood of statutory recognition. Meeting all three Census Town criteria in 2001 raises the probability of obtaining statutory recognition by 7.1 percentage points in our preferred local specification, with a first-stage F-statistic of 18.05. Using CT eligibility as an instrument for statutory town status, we identify local average treatment effects on public goods provision: ULB status increases the number of government primary schools by 13.86 per settlement, middle schools by 7.72, and secondary schools by 4.89. We also find positive effects on health and financial infrastructure: statutory recognition increases hospitals by 2.53, family welfare centers by 3.00, cooperative banks by 4.09, and agricultural credit societies by 2.84. Community amenities also improve, with positive estimates for public libraries and reading rooms, while sports infrastructure declines by 5.71 facilities, consistent with land reallocation in formalizing settlements¹. Results are robust to bandwidth choices, and robustness checks support the validity of the design. We find no evidence of manipulation around the thresholds and detect balance in baseline characteristics². The

¹Due to the small magnitude of the first-stage coefficient, the IV estimates scale reduced-form effects by a relatively large factor. The corresponding reduced-form estimates are directionally consistent and reported in the appendix.

²Our final regressions control for any remaining differences.

estimated effects also survive corrections for multiple hypothesis testing: summary indices constructed for each outcome family are all statistically significant, and the per-outcome reduced-form effects retain significance after Romano–Wolf and sharpened false-discovery-rate adjustments. The findings are not driven by any single census threshold—eligibility is identified by the *joint* satisfaction of all three criteria set by the Indian Census.

Our work builds on a growing literature that examines how local institutional capacity and governance shape development outcomes in low- and middle-income countries. [Besley and Persson \(2010\)](#) emphasize that investments in fiscal and legal capacity are key determinants of long-run development, and that variations in local administrative effectiveness can shape the quality of service delivery. Relatedly, [Gadenne \(2017\)](#) shows that greater fiscal autonomy can improve accountability and service delivery when citizens can observe how funds are used, while [Burgess et al. \(2015\)](#) highlights how political incentives affect the allocation of public goods across regions. We contribute to this literature by providing causal evidence on a different but complementary margin: the formal transition of villages to local urban governance status. Rather than focusing on local fiscal resources, we study how the administrative act of granting municipal status itself changes public goods provision. By leveraging the multi-threshold eligibility criteria for Census Town classification as a quasi-experiment, we isolate exogenous variation in the likelihood of statutory recognition and estimate its local average treatment effect on infrastructure outcomes. This approach provides new evidence on how the extension of local urban governance status affects provision of local public goods in urbanizing economies.

This paper contributes to the literature via two main fronts. First, it adds to the literature by quantifying the effect of local urban governance status on local public goods provision in large developing economies. Although [Denis and Marius-Gnanou \(2010\)](#), [Pradhan \(2013\)](#) and [Tewari \(2020\)](#) have noted that many Census Towns remain under rural governance despite meeting Census’s urban criteria, our study is the first to utilize a quasi-experimental setting to estimate the loss in public goods provision for these areas that are functionally

urban but rurally governed. Second, by leveraging the multi-threshold criteria for Census Town classification, our paper uses a novel instrument and highlights the potential for using multi-dimensional cutoffs in a fuzzy RD design setting. This approach can be applied to other contexts where policy eligibility is determined by multiple threshold-based criteria, allowing researchers to exploit variation in the likelihood of treatment assignment while accounting for the complexity of multi-dimensional eligibility rules. More broadly, our findings have implications for urban policy in rapidly urbanizing contexts, suggesting that timely municipalization of emerging urban areas can alter the provision of public goods.

2 Background & Data

2.1 Institutional Background

India’s urban classification rests on two pillars. The first consists of settlements officially designated as urban by state governments via notification under municipal law. These include municipal corporations, municipalities, and nagar panchayats³, and are each known as *Statutory Towns* (STs). The second consists of *Census Towns* (CTs), which are settlements that satisfy Census thresholds: total population at least 5,000; population density of at least 400 persons per km²; and at least 75 percent of the male main workforce in non-agricultural activities. CTs are urban for statistical purposes but may continue under rural governance unless notified as STs by the state governments. Crossing the CT thresholds increases the chance of statutory recognition but *does not guarantee it* as states differ in municipalization policy and timing. Some states have proactive policies to incorporate new urbanizing areas, while others delay or avoid creating new ULBs due to budgetary or political considerations.

³A type of town council for smaller urban areas.

2.2 Local Governance in India

The Nagarpalika Act, which was the 74th Amendment to the Indian Constitution enacted in 1992, provides the legal framework for urban local governance in India ([Ministry of Home Affairs, Government of India, 2024](#)). This constitutional amendment is supported by state-level municipal acts, such as [Government of Goa, 1968](#); [Government of Gujarat, 1963](#); [Government of Karnataka, 1964](#); [Government of Maharashtra, 1965](#); [Government of Telangana, 2019](#) etc., that outline the specific procedures for forming Urban Local Bodies (ULBs). The entire process, from initial identification to operational ULB, can take several years and is subject to state-specific variations in policy, political will, and administrative capacity. Some states proactively municipalize urbanizing settlements, while others delay recognition due to fiscal or political constraints, leading to the phenomenon of Census Towns experiencing an *urban governance gap*—settlements that are functionally urban but remain under rural governance.

2.2.1 What comes with ULB recognition?

When a settlement receives statutory recognition as an Urban Local Body (ULB), it undergoes a fundamental transformation in governance structure, fiscal arrangements, and administrative responsibilities. The 74th Constitutional Amendment Act of 1992 mandates that ULBs assume control over 18 functions listed in the Twelfth Schedule, including urban planning, regulation of land use, roads and bridges, water supply, public health and sanitation, fire services, urban poverty alleviation, and provision of urban amenities ([Ministry of Home Affairs, Government of India, 2024](#)). Crucially, ULB recognition brings changes in three key dimensions:

Governance and Administrative Structure: Statutory towns transition from rural governance under gram panchayats to elected municipal councils or corporations. This shift creates a dedicated urban administrative apparatus with specialized departments for engineering, health, education, and revenue collection. ULBs are headed by elected mayors or municipal chairpersons, with ward-level representation ensuring political accountability for

urban service delivery.

Fiscal Autonomy and Revenue Sources: ULBs gain the authority to levy and collect municipal taxes, most notably property taxes, as well as user charges for water, sanitation, and other services. They also become eligible for devolved funds from state finance commissions and centrally-sponsored urban schemes such as the Jawaharlal Nehru National Urban Renewal Mission (JNNURM) and later the Smart Cities Mission and AMRUT (Atal Mission for Rejuvenation and Urban Transformation). While many ULBs remain fiscally dependent on state transfers, statutory recognition opens access to dedicated urban funding streams unavailable to rural settlements.

Service Delivery Mandates: Upon municipalization, ULBs inherit responsibility for providing and maintaining urban infrastructure and services. This includes establishing and operating primary schools, health dispensaries, water supply systems, sewerage and drainage networks, street lighting, solid waste management, and fire protection services.

However, the practical impact of ULB recognition is heterogeneous. Well-functioning municipalities in states with strong urban governance traditions may deliver substantially better services, while newly formed ULBs in resource-constrained states may struggle with capacity limitations, inadequate transfers, and weak revenue collection.

2.2.2 How does Census Town classification affect statutory recognition?

Although Census Town (CT) classification and statutory recognition are distinct processes that are built on different criteria and governed by different authorities, there is a strong relationship between the two and one often inspires the other ([Tewari, 2020](#); [Roy and Pradhan, 2018](#)). Meeting the CT criteria signals that a settlement has urban characteristics and may warrant municipal governance. When a settlement crosses the CT thresholds, it draws attention from state urban development authorities and policymakers, who may then consider it for notification as a statutory town. The CT designation provides an objective benchmark indicating that the settlement has reached a level of population size, density,

and economic activity consistent with urban areas. This can prompt state governments to evaluate whether the settlement is ready for municipal governance and whether it can sustain the administrative and fiscal responsibilities of a ULB. In fact, the central government also monitors CTs and encourages states to municipalize them, as exhorted by [Ministry of Housing and Urban Affairs, Government of India \(2016\)](#) where the Ministry of Housing and Urban Affairs (MoHUA) formally urged states to consider CTs for statutory recognition.

2.3 Data Sources and Construction

Using SHRUG data platform⁴, we collect data at the village and town level⁵ for our study ([Asher, Novosad et al., 2019](#)). We also collect state-level town directories from the Government of India’s Open Government Data (OGD) platform and list of Statutory Towns (STs) from National Housing Bank’s (NHB) website. We highlight the main datasets used in our analysis:

- **Population Census Data (2001 & 2011):** We draw on comprehensive census data for all Indian settlements across two decennial rounds. The Primary Census Abstract (PCA) provides basic demographic characteristics including total population, gender and age composition, Scheduled Caste and Scheduled Tribe populations, literacy rates, and detailed workforce composition. These data allow us to construct the three Census Town eligibility criteria: population size, population density, and share of male main workers in non-agricultural activities.
- **Village and Town Directories:** We obtain infrastructure and service provision data from SHRUG’s harmonized Census Village and Town Directories for 2001 and 2011. These include measures of education facilities such as primary through senior secondary schools, health infrastructure such as hospitals, dispensaries, primary health centers, family welfare centers, financial access variables such as cooperative banks, agricultural

⁴version 2.1 was available at the time of our study.

⁵This is the most granular level at which information is available and is identified using SHRUG IDs.

credit societies, and various community amenities. When definitions differ between the Village and Town Directories, we create harmonized outcome variables to ensure comparability across rural and urban settlements⁶.

- **Statutory Town Lists:** We obtain comprehensive lists of all statutorily recognized urban local bodies from the National Housing Bank and state-level census town directories from the Government of India’s Open Government Data (OGD) platform.

Our final sample consists of over 500,000 settlements in India, which includes most inhabited villages and towns included in the 2001 and 2011 Census. Of these, over 3,700 have statutory urban status i.e. municipalities of some form as of 2011, while the remaining are governed as rural villages. However, many of those rural-governed settlements are urban in character. In total, about 9,000 settlements meet the Census definition of urban. This implies that roughly 2% of all Indian settlements were considered urban in 2011 by this criterion, and the rest were rural villages. These figures highlight that India’s urban population is concentrated in a relatively small number of settlements – which tend to be much larger on average than the multitude of villages.

2.4 Outcome Selection: Mapping Public Goods to Urban Local Body Mandates

Our choice of outcomes is guided by two principles rather than data availability alone. First, we focus on local public goods that fall within the service-delivery mandate that the 74th Constitutional Amendment assigns to Urban Local Bodies. The Twelfth Schedule charges ULBs with functions including education, public health and sanitation, urban planning, and the provision of urban amenities such as parks, playgrounds, and cultural facilities ([Ministry of Home Affairs, Government of India, 2024](#)). Second, we restrict attention to outcomes that

⁶The variable crosswalk and harmonization rules are documented in the codebook supplied with the replication package.

are measured consistently across the *2011 Village Directory* and *Town Directory*, so that rural-governed Census Towns and statutory towns are compared on a common definition⁷.

We group the outcomes into four distinct categories that reflect different dimensions of local public goods provision:

- **Education:** We focus on government primary, middle, and secondary schools. The provision and maintenance of schools is an explicit ULB function, and the rural baseline—education under the gram panchayat—offers a natural comparison. We use government schools to isolate publicly provided infrastructure.
- **Health:** We focus on hospitals and family welfare centers. Public health, sanitation, and the operation of dispensaries, maternal healthcare and child-health facilities are core Twelfth-Schedule responsibilities, making family welfare centers central to India’s public health system (Dhingra and Dutta, 2011).
- **Community amenities:** We focus on public libraries, reading rooms, cinema halls, and sports infrastructure. These map directly to the Twelfth-Schedule mandates to promote “cultural, educational and aesthetic aspects” and to provide “urban amenities and facilities such as parks, gardens, [and] playgrounds.” We flag sports infrastructure as the least cleanly harmonized of our outcomes across the Village and Town Directories⁸.
- **Financial access:** We focus on cooperative banks and agricultural credit societies. Unlike the categories above, the establishment of banks is *not* a direct ULB function. We include these outcomes mainly because of this distinction: any effect on financial access could operate indirectly, through the administrative upgrade itself—statutory recognition can signal permanence and creditworthiness to financial institutions and raise local demand. Contrasting these “induced” outcomes with the directly provided

⁷The harmonization is detailed in the codebook supplied with the replication package.

⁸See the codebook supplied with the replication package for the full variable crosswalk.

public goods above is therefore informative about the channels through which urban governance operates⁹.

2.5 Running Variables and Treatment Assignment

The key running variables for our analysis are the three Census Town eligibility criteria: population size, population density, and share of male main workers in non-agricultural activities. We construct these variables using the 2001 Census data for each settlement. Specifically, we consider the following definitions for the running variables¹⁰:

- **Population Size (P):** Total population of the settlement as recorded in the 2001 Census.
- **Population Density (D):** Total population in 2001 divided by the area of the settlement in square kilometers, computed from the area measures provided in the Village and Town Directories.
- **Non-Agricultural Share (N):** The proportion of male main workers engaged in non-agricultural activities in 2001, calculated as the number of male main workers in non-agricultural sectors divided by the total number of male main workers in the settlement.

We then define the treatment variable for statutory recognition based on whether a settlement was officially designated as a statutory town (ST) in 2011. This is a binary indicator that takes the value of 1 if the settlement was recognized as an ST in 2011, and 0 otherwise. The treatment assignment is not deterministic based on the running variables alone, as some settlements that meet the Census Town criteria may not receive statutory recognition due to state-level policy decisions, while others that do not meet the criteria may be recognized due to historical or political reasons. Therefore, we use a fuzzy regression

⁹See Section 4 for more details.

¹⁰Density plots of these variables are presented in the Appendix.

discontinuity design that exploits the discontinuities in the probability of receiving statutory recognition at the thresholds defined by the running variables to identify the causal effect of local urban governance status on local public goods provision.

2.6 Summary Statistics

2.6.1 Urban/Rural Classification vs Statutory Recognition

We define statutory recognition using the state-level files available on Indian government’s Open Government Data (OGD) platform¹¹. These files report the list of statutory towns for each state as of 2011. We merge these files with the PCA data using state and settlement names to create an indicator for statutory recognition.

Table 1: Statutory Recognition and Urban Thresholds

Panel A: Population $\geq 5,000$		
	Pop $< 5,000$	Pop $\geq 5,000$
Not Statutory	485,580	17,509
Statutory	159	3,570
Panel B: Density ≥ 400 per km²		
	Density < 400	Density ≥ 400
Not Statutory	308,943	194,146
Statutory	169	3,560
Panel C: Non-Agricultural Male Workers $\geq 75\%$		
	Non-Ag $< 75\%$	Non-Ag $\geq 75\%$
Not Statutory	467,234	35,855
Statutory	1,056	2,673
Panel D: All Three Thresholds Combined		
	Did not Meet Thresholds	Met Thresholds
Not Statutory	499,708	3,381
Statutory	1,194	2,535

Table 1 provides a descriptive sense of this process by cross-tabulating threshold eligibility in 2001 and statutory status by 2011. Several patterns stand out. First, 17,509 settlements

¹¹State-level files were collected via <https://data.gov.in/>.

had populations above 5,000 yet were not statutory towns by 2011, compared to 3,570 settlements that were both above 5,000 and statutorily recognized (Panel A of Table 1). A similar discrepancy is seen for the density criterion (Panel B), where 194,146 settlements had density above 400 per km² in 2001 but were not statutory towns by 2011, while only 3,560 settlements met the density criterion and had statutory town status. For the non-agricultural workforce criterion (Panel C), 35,855 settlements had at least 75% non-agricultural male workers but were not statutory towns, compared to 2,673 that met this criterion in 2001 and were statutory towns by 2011. Second, looking at the combined criteria (Panel D), a total of 5,916 settlements met all three urban criteria in 2001 (3,381 not statutory + 2,535 statutory). However, only 2,535 of these were actually recognized as statutory towns by 2011, while 3,381 settlements met the criteria but remained governed as rural. Conversely, there were 1,194 statutory towns that did not meet all the Census criteria. This highlights that satisfying the Census definition is not sufficient for urban governance as political implementation by states is crucial. It also underscores the quasi-random aspect of our design: among settlements around the thresholds, some received statutory recognition by 2011 and others did not, not purely based on merit but partly due to state-specific policies and timing. Nevertheless, as we will show in Section 3, crossing the thresholds greatly increases the likelihood of statutory recognition, providing the basis for our fuzzy regression discontinuity approach.

2.6.2 Treatment Assignment and Treatment Status

Table 2 presents the distribution of settlements by Census Town (CT) classification in 2001 and statutory town (ST) status by 2011. Panel A shows the full sample of all settlements, while Panel B restricts to settlements close to the CT thresholds.

Several patterns emerge from Table 2. In the full sample (Panel A), the vast majority of settlements (98.6%) neither met the CT criteria in 2001 nor received statutory recognition by 2011. Among the 5,765 settlements that met the CT thresholds in 2001, only 2,542 (44%) gained statutory status by 2011, while 3,223 (56%) remained under rural governance despite

Table 2: Treatment Assignment and Status

Panel A: Full Sample			
CT in 2001	Statutory by 2011	N	Share (%)
No	No	495,215	98.61
No	Yes	1,199	0.24
Yes	No	3,223	0.64
Yes	Yes	2,542	0.51
Total		502,179	100.00

Panel B: Close to Threshold Sample			
CT in 2001	Statutory by 2011	N	Share (%)
No	No	36,726	98.86
No	Yes	100	0.27
Yes	No	297	0.80
Yes	Yes	28	0.08
Total		37,151	100.00

Notes: “CT in 2001” indicates whether a settlement met all three Census Town criteria (population $\geq 5,000$, density ≥ 400 per km², non-agricultural workers $\geq 75\%$) in 2001. “Statutory by 2011” indicates whether the settlement had statutory town status in 2011. Panel B restricts to settlements within the bandwidth around the CT thresholds.

being functionally urban. This substantial gap between eligibility and recognition illustrates the discretionary nature of state municipalization policy. Conversely, 1,199 settlements that did not meet the CT criteria nevertheless received statutory recognition, typically reflecting historical municipal status or special circumstances.

The close-to-threshold sample (Panel B) reveals similar patterns on a smaller scale, with 297 settlements meeting CT criteria but not receiving statutory status, and 100 settlements gaining statutory status despite not meeting the criteria. This variation around the thresholds provides the identifying variation for our fuzzy regression discontinuity design, where crossing the CT thresholds substantially increases but does not guarantee statutory recognition.¹²

¹²For descriptive view of outcomes by statutory status in the global and local samples, see Appendix Table 1.

3 Empirical Strategy

In this section, we lay out the identification strategy. The goal is to estimate the causal effect of statutory recognition and ULB status on settlement outcomes.

3.1 Quasi-random variation from Census Towns

In this section, we show that Census Town (CT) designation leads to a discontinuous increase in the probability of statutory recognition at the thresholds. Furthermore, becoming a CT does not directly change governance status as the decision to grant a ULB is made by the state government. So theoretically, we should not expect a direct effect of CT designation on local outcomes through the local urban governance status channel unless the settlement is also statutorily recognized. One could argue that CT may be capturing the differences in agglomeration economies or other unobserved characteristics that may be correlated with local outcomes. However, close to the thresholds, these characteristics should be similar on either side of the cutoff. For a country of 1.5 billion people, a settlement with a population of 4,900 and 5,100, or density of 390 and 410, is not likely to be systematically different in terms of agglomeration economies. Furthermore, even though we cannot rule out differences in unobserved characteristics, we can test for balance in observed characteristics¹³ and run placebo tests at pseudo-thresholds to check if there are any discontinuities in outcomes at these points. First, we show that there is a strong first-stage relationship between meeting the CT thresholds and statutory recognition. We then use this quasi-random variation to instrument for statutory recognition and identify the effect of statutory recognition on local outcomes.

Below we describe how we implement this empirical strategy in practice. Let P_i , D_i , and N_i denote the population, density (per km²), and non-agricultural employment share of settlement i . Define three binary indicators for meeting the CT thresholds:

¹³See Appendix Table 2.

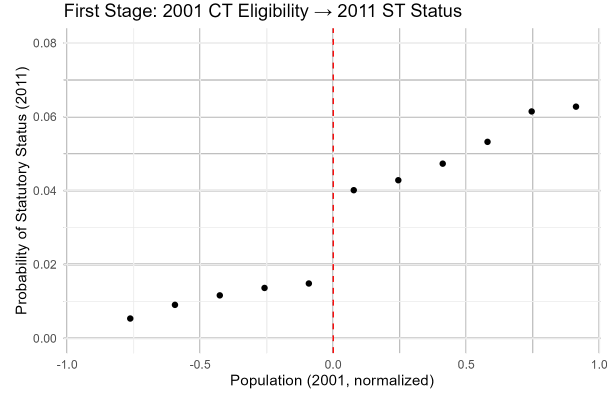
$$\begin{aligned}
Z_{P,i} &= 1 \left\{ \frac{P_i}{5000} - 1 \geq 0 \right\} \\
Z_{D,i} &= 1 \left\{ \frac{D_i}{400} - 1 \geq 0 \right\} \\
Z_{N,i} &= 1 \left\{ \frac{N_i}{0.75} - 1 \geq 0 \right\}
\end{aligned}$$

We define CT eligibility as $Z_i = Z_{P,i} \times Z_{D,i} \times Z_{N,i}$, which equals 1 if settlement i meets all three thresholds and 0 otherwise, which we use as an instrument for statutory recognition. The idea is that settlements that meet the CT thresholds are more likely to receive statutory recognition, but the decision to grant statutory recognition is not deterministic based on the CT thresholds alone. Therefore, we can use the variation in CT eligibility to identify the causal effect of statutory recognition on local outcomes.

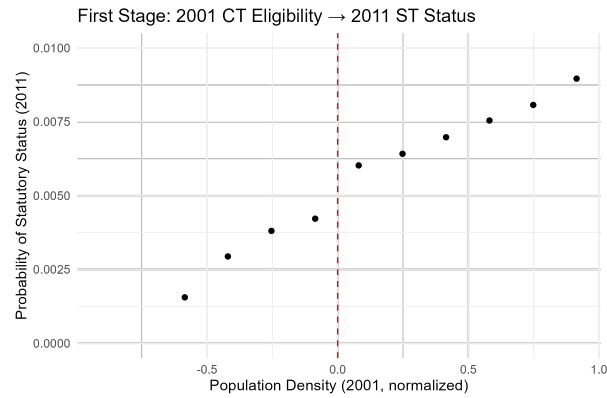
Next, let ST_i be an indicator for statutory recognition by 2011 i.e. whether settlement i gets an Urban Local Body (ULB) or not, by 2011. Then, our object of interest is the probability of statutory recognition $P(ST_i = 1 | Z_i, X_i)$, where X_i is a vector of controls. Our data enables us to estimate the following first stage regression:

$$ST_i = \pi_0 + \pi_1 Z_i + X_i' \beta + \delta_{s(i)} + \varepsilon_i \quad (1)$$

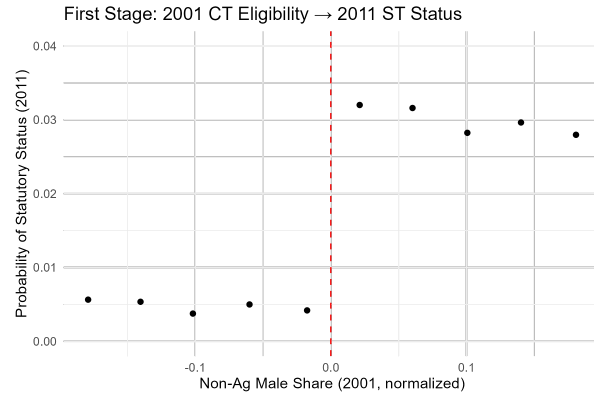
where Z_i is CT eligibility in 2001 (instrument), X_i is a vector of smooth functions of the running variables and other controls (e.g. $Z_{P,i}, Z_{D,i}, Z_{N,i}$, literacy, caste shares), and $\delta_{s(i)}$ are district-level fixed effects.



(A) Population threshold



(B) Density threshold



(C) Non-agricultural share threshold

Figure 1: First-stage Discontinuities and Statutory Recognition

Notes: Each panel shows a binned scatter plot of the predicted probability of statutory recognition in 2011 against one of the three normalized running variables from 2001. Predicted probabilities are fitted values from the global first-stage OLS regression of statutory status on Census Town eligibility, controls, and district fixed effects (Equation 1). The dashed red line marks the normalized threshold.

Figure 1 presents graphically the first stage of our fuzzy RD design explained in equation 1 close to the thresholds¹⁴. The figure shows binned scatter plots of the probability of statutory status in 2011 against each running variable in 2001. Sub-figure (A) of Figure 1 shows the relationship between population and statutory recognition. Probability of statutory recognition is positively related to population, which is expected since an increase in population signifies a larger settlement that is more likely to be recognized as a town. Despite this positive relationship, there is a clear discontinuity in the probability of statutory recognition at the population threshold of 0 after normalization. This discontinuity indicates that settlements that just meet the population threshold are significantly more likely to receive statutory recognition than those that just miss it. Sub-figure (B) of Figure 1 shows the relationship between density and statutory recognition. Similar to population, there is a positive relationship between density and statutory recognition, but the discontinuity at the density threshold, while present, is less pronounced than for population. Sub-figure (C) of Figure 1 shows the relationship between non-agricultural share and statutory recognition. The relationship is flat, but there is a clear discontinuity at the non-agricultural share threshold. Settlements that just meet the non-agricultural share threshold are more likely to receive statutory recognition than those that just miss it. Overall, these figures provide evidence of a strong first-stage relationship between meeting the CT thresholds and statutory recognition, which supports our identification strategy for estimating the causal effect of statutory recognition on local outcomes. In Section 4.1, we present a table that reports the first-stage coefficient and F-statistics for global and local estimates.

3.2 Fuzzy RD with multiple running variables

India's CT thresholds create jointly-binding rules on three dimensions: population $P \geq 5,000$, density $D \geq 400$ per km², and male main non-agricultural share $N \geq 0.75$. Define running variables as normalized distances to the cutoffs: $r_P = P/5000 - 1$, $r_D = D/400 - 1$,

¹⁴We present results up to 100% away from the thresholds.

and $r_N = N/0.75 - 1$. Following the logic of multi-cutoff RD, we focus on observations near the cutoffs and develop a frontier using a soft-min functional specification.

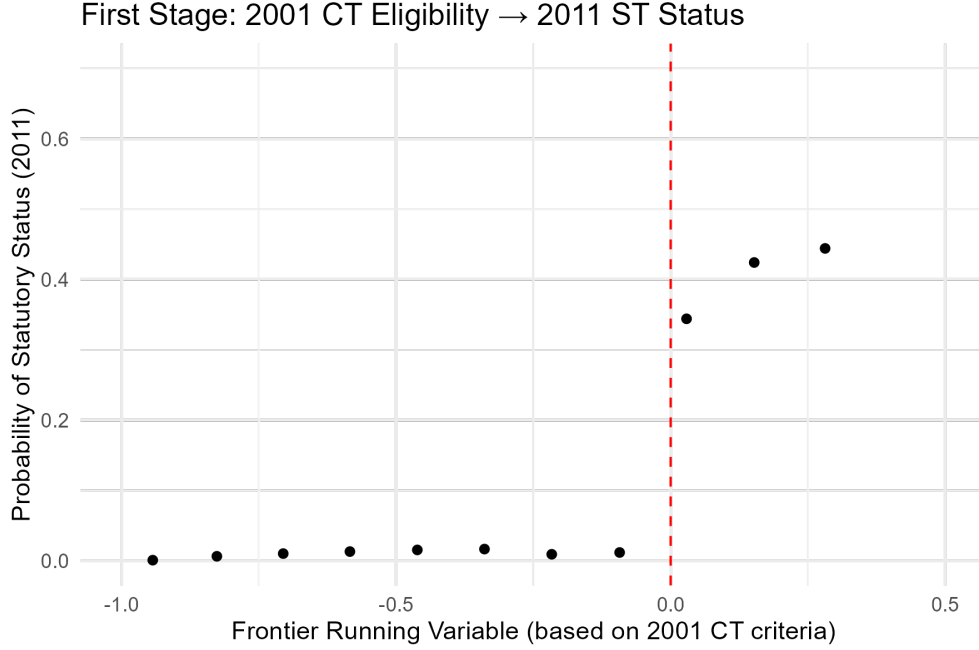


Figure 2: Probability of Statutory Recognition by Frontier Distance

Notes: Binned scatter plot of the predicted probability of statutory recognition in 2011 against the soft-min frontier running variable constructed from the three 2001 Census Town criteria. The frontier variable is defined as $r_i^* = -\frac{1}{\alpha} \ln(\exp(-\alpha r_P) + \exp(-\alpha r_D) + \exp(-\alpha r_N))$ with $\alpha = 10$, which smoothly approximates the minimum of the three normalized distances. Negative values indicate that a settlement falls short of at least one threshold; positive values indicate that all three thresholds are met. Each dot represents the mean predicted probability within an equal-width bin. Predicted probabilities are fitted values from the global first-stage regression¹⁵ (Equation 1).

Figure 2 plots the predicted probability of statutory recognition against the frontier running variable, which collapses the three Census Town dimensions into a single scalar. The figure reveals a clear discontinuity at the eligibility frontier, near the zero on the horizontal axis. To the left of the cutoff, where settlements fall short of at least one threshold, the predicted probability of receiving a municipal body is close to zero and hovers near 1%.

Immediately to the right, where settlements satisfy all three criteria, the probability jumps sharply to approximately 35%, rising further to above 40% for settlements that comfortably exceed the thresholds. This pattern is precisely what the fuzzy RD design requires: a large, discrete change in the probability of treatment at the eligibility boundary, embedded within an otherwise smooth relationship. The magnitude of the jump is substantial and visually unambiguous, providing strong graphical evidence that Census Town eligibility is a predictor of statutory recognition. This reinforces the relevance condition of our instrument and is consistent with the formal first-stage F-statistics reported in Table 3.

The contrast between Figure 2 and the individual-threshold plots in Figure 1 is also informative. When projected onto any single running variable, the discontinuity exists but appears modest because most of the variation in eligibility is driven by the joint interaction of all three criteria. The frontier variable captures this joint variation and motivates our use of a multi-dimensional instrument defined by the intersection of all three thresholds, rather than relying on any single cutoff.

3.3 Estimation

We estimate the effect of statutory recognition on local public goods using the local or close-to-threshold sample. Let Y_i denote an outcome for settlement i measured in 2011—in our specifications, the *count or number of a given public good* in that settlement (e.g. number of government primary schools, hospitals, cooperative banks). The **reduced-form** equation relates this outcome directly to Census Town eligibility:

$$Y_i = \gamma_0 + \gamma_1 Z_i + X_i' \gamma + \delta_{s(i)} + u_i, \quad (2)$$

where Z_i is the CT-eligibility instrument, X_i is the control vector, and $\delta_{s(i)}$ are district fixed effects. The reduced-form or intent-to-treat (ITT) coefficient γ_1 measures the effect of crossing the eligibility frontier on the outcome. Instrumenting statutory recognition ST_i

with Z_i , the second-stage (2SLS) equation is

$$Y_i = \beta_0 + \beta_1 ST_i + X_i' \beta + \delta_{s(i)} + \varepsilon_i, \quad (3)$$

where β_1 is the local average treatment effect of statutory (ULB) recognition. In the just-identified case, $\beta_1 = \gamma_1/\pi_1$, the ratio of the reduced-form effect to the first-stage coefficient from Equation (1)¹⁶.

4 Results

4.1 First stage: Effects on Statutory Recognition

Table 3: First-stage: Effect of Census-Town Eligibility on Statutory Status

	Global	Local
<i>Dependent variable:</i> Statutory town = 1		
CT eligibility in 2001	0.431*** (0.0157)	0.071*** (0.0167)
Controls	Yes	Yes
District fixed effects	Yes	Yes
SEs clustered at district level	Yes	Yes
First-stage F -statistic	749.43	18.05
Partial R^2 of instrument	0.2522	0.0127
Adj. R^2	0.3493	0.1007
Observations	502,179	37,151
Districts (FE groups)	619	558

Notes: First-stage estimates of statutory recognition on the census-threshold instrument with controls and district fixed effects. Controls include population, density, non-agricultural male workforce share, literacy rate, and caste composition. Standard errors are clustered by district (in parentheses). “Local” restricts to settlements near the 2001 census thresholds (population $\approx 5,000$, density ≈ 400 , non-agricultural male workforce share ≈ 0.75). The single-IV first-stage F is t^2 from the clustered specification.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The first-stage results in Table 3 reveal important patterns regarding the strength and

¹⁶Standard errors are clustered at the district level throughout.

validity of our instrumental variable strategy. In the global sample, the instrument exhibits substantial explanatory power with a coefficient of 0.431 and a first-stage F-statistic of 749.43, well exceeding conventional thresholds for weak instrument concerns. However, this strong statistical relationship may be misleading for causal identification purposes. The global specification, while statistically powerful, is potentially vulnerable to endogeneity. Settlements across the entire population and density spectrum may differ systematically in ways that correlate with both census threshold compliance and subsequent outcomes. For instance, larger settlements may have greater political influence or administrative capacity that simultaneously makes them more likely to meet census criteria and more likely to attract infrastructure investment, regardless of statutory status.

In contrast, the local specification of restricting analysis to settlements near the census cutoffs¹⁷ provides more credible identification despite the smaller coefficient magnitude (0.071) and relatively lower F-statistic (18.05). This specification focuses on settlements that are observationally similar except for small differences in census characteristics that push them just above or below the thresholds. The lower adjusted R^2 (0.1007) and partial R^2 of instrument (0.0127) indicate that we are capturing variation that is more plausibly exogenous, as settlements just above and below the cutoffs should be similar in unobservable characteristics in the absence of treatment, as long as no manipulation occurs.

The F-statistic of 18.05 for our local specification is above the conventional threshold of 16.38 for strong instruments in the just-identified case (Staiger and Stock, 1997) and well above the critical value of 8.96 for 15% maximal IV bias (Stock and Yogo, 2002), suggesting the instrument maintains reasonable strength for our identification strategy.

4.2 Main effects on outcomes

We next present 2SLS estimates of the effect of statutory recognition on settlement outcomes. We group outcomes into categories most relevant to local public goods and services:

¹⁷We use the following restrictions around the thresholds: (i) population ± 5000 (ii) density ± 400 (iii) non-ag male main work share ± 0.2 . Appendix Table 3 presents bandwidth-sensitivity results.

education, health, financial access, and community infrastructure¹⁸.

Table 4: Effect of Local Urban Governance Status on School Provision

	Dependent variable: Number of government schools		
	Primary	Middle	Secondary
Effect of ULB status	13.86*** (4.00)	7.72*** (2.25)	4.89*** (1.30)
Controls	Yes	Yes	Yes
District FE	Yes	Yes	Yes
Observations	37,143	37,099	37,148

Notes: Table reports two-stage least squares estimates where the endogenous regressor is statutory town status (ULB = 1) and the excluded instrument is Census Town eligibility in 2001. The local sample restricts to settlements near the 2001 census thresholds (population $\pm 5,000$; density ± 400 ; non-agricultural male share ± 0.20). All specifications include controls for log population, log density, non-agricultural male workforce share, literacy rate, main worker share, and caste composition (SC and ST shares), as well as district fixed effects. Robust standard errors clustered at the district level are reported in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Education. Table 4 presents the estimated effects of statutory recognition on educational infrastructure provision. The results demonstrate substantial positive impacts across all levels of schooling. Statutory recognition leads to an increase of 13.86 additional primary schools, 7.72 additional middle schools, and 4.89 additional secondary schools. All estimates are statistically significant at the 1% level, with robust standard errors clustered at the district level. The largest absolute effect occurs at the primary level, consistent with primary education being a foundational service that local governments can expand most readily. Primary schools are also typically smaller in size and easier to open than middle or secondary schools, facilitating rapid increases in provision following statutory recognition. As we move up the educational ladder, the estimated effects become smaller, which is plausible given the higher capital requirements, larger catchment areas, and greater regulatory demands associated with middle and secondary schools. Even so, the fact that all three levels

¹⁸The reduced-form estimates provide the numerator of the Wald ratio underlying our IV specification and show that crossing the CT eligibility frontier improves public goods provision across the same outcome categories. The full reduced-form table is reported in Appendix Table 4.

show significant positive effects suggests that ULB status facilitates broad improvements in educational infrastructure rather than a narrow shift at only one school level.

Table 5: Effect of Local Urban Governance Status on Hospital Provision

	Dependent variable	
	All hospitals	Family welfare centers
Effect of ULB status	2.53*** (0.69)	3.00*** (0.88)
Controls	Yes	Yes
District FE	Yes	Yes
Observations	37,091	37,077

Notes: Two-stage least squares estimates where the endogenous regressor is statutory town status ($ULB = 1$) and the excluded instrument is Census Town eligibility in 2001. The local sample restricts to settlements near the 2001 census thresholds (population $\pm 5,000$; density ± 400 ; non-agricultural male share ± 0.20). All specifications include controls for log population, log density, non-agricultural male workforce share, literacy rate, main worker share, and caste composition (SC and ST shares), as well as district fixed effects. Robust standard errors clustered at the district level are reported in parentheses.

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Health. Table 5 presents the estimated effects of local urban governance status on health infrastructure provision. The results demonstrate substantial positive impacts across both health facilities. Local urban governance status leads to an increase of 2.53 additional hospitals and 3.00 additional family welfare centers. Both estimates are statistically significant at the 1% level, with robust standard errors clustered at the district level. The positive effects on both general hospitals and family welfare centers suggest that local urban governance status facilitates improvements in core health infrastructure. The somewhat larger effect on family welfare centers is noteworthy because these facilities play a central role in maternal and child health, family planning, and preventive care within India’s public health system (Dhingra and Dutta, 2011). This pattern is also plausible given that family welfare centers typically require less capital and can be expanded more quickly than full-service hospitals.

Table 6: Effect of Local Urban Governance Status on Financial Access

	Dependent variable	
	Cooperative banks	Agricultural credit societies
Effect of ULB status	4.09*** (0.97)	2.84*** (1.01)
Controls	Yes	Yes
District FE	Yes	Yes
Observations	37,089	37,093

Notes: Two-stage least squares estimates where the endogenous regressor is statutory town status ($ULB = 1$) and the excluded instrument is Census Town eligibility in 2001. The local sample restricts to settlements near the 2001 census thresholds (population $\pm 5,000$; density ± 400 ; non-agricultural male share ± 0.20). All specifications include controls for log population, log density, non-agricultural male workforce share, literacy rate, main worker share, and caste composition (SC and ST shares), as well as district fixed effects. Robust standard errors clustered at the district level are reported in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Financial Access. Table 6 presents the estimated effects of local urban governance status on financial infrastructure provision. The results demonstrate substantial positive impacts across both types of financial institutions. Local urban governance status leads to an increase of 4.09 additional cooperative banks and 2.84 additional agricultural credit societies, both statistically significant at the 1% level with robust standard errors clustered at the district level. The positive effects across both institution types indicate that municipalization enhances multiple forms of financial intermediation serving different community needs. Cooperative banks can support broader commercial and household financial activity, while agricultural credit societies remain relevant in peri-urban settlements where agricultural livelihoods persist alongside urbanization. The expansion of financial infrastructure may therefore reflect both greater demand in newly recognized towns and the possibility that local urban governance status signals administrative capacity and development potential to formal financial institutions (Burgess and Pande, 2005).

Community Infrastructure. Table 7 presents the estimated effects of local urban governance status on community infrastructure and access. The results show positive effects

Table 7: Effect of Local Urban Governance Status on Community Infrastructure

	Dependent variable			
	Public libraries	Public reading rooms	Cinema halls	Sports infrastructure
Effect of ULB status	1.05** (0.50)	1.44** (0.63)	0.79 (0.49)	-5.71*** (1.48)
Controls	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Observations	37,147	37,147	37,146	37,146

Notes: Two-stage least squares estimates where the endogenous regressor is statutory town status (ULB = 1) and the excluded instrument is Census Town eligibility in 2001. The local sample restricts to settlements near the 2001 census thresholds (population $\pm 5,000$; density ± 400 ; non-agricultural male share ± 0.20). All specifications include controls for log population, log density, non-agricultural male workforce share, literacy rate, main worker share, and caste composition (SC and ST shares), as well as district fixed effects. Robust standard errors clustered at the district level are reported in parentheses.

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

on several cultural and educational amenities. Local urban governance status leads to a statistically significant increase of 1.05 public libraries and 1.44 public reading rooms. The estimate for cinema halls is positive at 0.79 but imprecisely estimated. By contrast, sports infrastructure declines by 5.71 facilities. This negative effect is consistent with spatial constraints and land-use tradeoffs in urban development: as settlements formalize into statutory towns, open spaces previously used for sports facilities may be repurposed for schools, health facilities, banking infrastructure, roads, or residential development. Overall, these findings suggest that formal statutory recognition may enhance access to community facilities, though the transition may involve restructuring of certain types of infrastructure, particularly those requiring substantial land area¹⁹.

4.3 Robustness Checks

4.3.1 Density Plots and McCrary Test for Manipulation

We conduct checks around the statutory recognition thresholds to assess the validity of our RD design. Specifically, we plot the density of the running variables and implement the

¹⁹For a comparison of these 2SLS estimates against global and local OLS, see Appendix Table 5.

McCrary (2008) test for manipulation of the running variable, which examines the continuity of the density of the running variable at the cutoff.

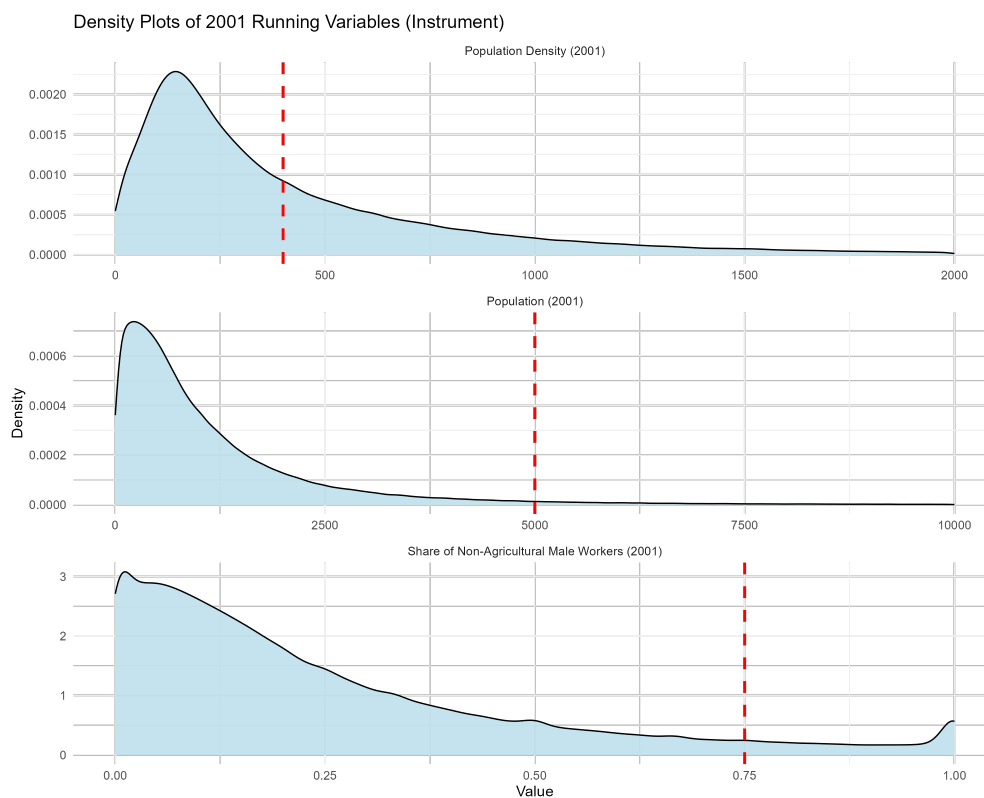


Figure 3: Density plots of CT-definition variables and thresholds

Figure 3 shows the density plots of the three running variables (population density, population, and share of non-agricultural male workers) around their respective cutoffs of 400 persons per km², 5,000 persons, and 0.75. The plots indicate that the density of the running variables is continuous and smooth around the cutoffs, with no clear signs of manipulation or strategic sorting.

Table 8: McCrary Density Tests at CT Thresholds

Running Variable	Cutoff	T-statistic	P-value
Population (2001)	5,000	0.754	0.451
Population Density (2001)	400	0.245	0.807
% Non-Agricultural Male Workers (2001)	0.75	−1.101	0.271

Notes: This table reports results from the [McCrary \(2008\)](#) manipulation test which examines whether there is a discontinuity in the density of the running variable at the statutory threshold. The T-statistic tests the null hypothesis of no discontinuity. All tests use a triangular kernel with bandwidth selection and jackknife variance estimation. P-values greater than 0.10 indicate no evidence of manipulation at conventional significance levels.

The [McCrary \(2008\)](#) density test formally assesses whether there is evidence of manipulation of the running variable around the cutoff. The test estimates the density of the running variable separately on each side of the threshold and tests whether there is a discontinuity in the density at the cutoff. A significant discontinuity would suggest strategic sorting around the threshold, which would violate the identifying assumptions of the RD design.

Table 8 presents the results of the [McCrary \(2008\)](#) test for each of our three running variables used to define census town status. For population in 2001 (cutoff at 5,000), the test statistic is 0.754 with a p-value of 0.451. For population density in 2001 (cutoff at 400 persons per km²), the test statistic is 0.245 with a p-value of 0.807. For the share of non-agricultural male workers (cutoff at 0.75), the test statistic is −1.101 with a p-value of 0.271. In all three cases, we fail to reject the null hypothesis of no discontinuity in the density at conventional significance levels. This provides reassuring evidence that there is no systematic manipulation of these running variables around the statutory thresholds.

4.3.2 Multiple Hypothesis Testing

Since we estimate effects on multiple outcomes, we guard against the possibility that some individual results are significant by chance. We pursue the two standard remedies in tandem ([Anderson, 2008](#)): aggregating outcomes into family-level summary indices, and adjusting the per-outcome p-values for multiplicity.

First, we construct an [Anderson \(2008\)](#) inverse-covariance-weighted (ICW) index for each of our four outcome families—education, health, financial access, and cultural amenities—by standardizing each component²⁰, weighting components by the inverse of their covariance matrix, and re-standardizing the final index. This collapses the many outcomes into four indices and gains power by exploiting the correlation among related outcomes. As suggested by results in Appendix Table 6, crossing the Census Town eligibility frontier raises the education index by 0.90 SD, the health index by 1.29 SD, the financial-access index by 0.90 SD, and the cultural-amenuities index by 0.32 SD; all four are statistically significant at the 5% significance level²¹.

Second, we adjust the per-outcome reduced-form p-values for multiple testing using the [Romano and Wolf \(2005\)](#) stepdown procedure which controls the familywise error rate while accounting for cross-outcome correlation via a district-level cluster bootstrap, and the [Anderson \(2008\)](#) sharpened two-stage false-discovery-rate q-values ([Benjamini, Krieger and Yekutieli, 2006](#)). Appendix Table 7 reports the unadjusted, Romano–Wolf, and sharpened q-values side by side. Every main outcome remains significant at the 5% level after the Romano–Wolf adjustment, with the exception of cinema halls. All eleven outcomes have sharpened FDR q-values below 0.012, rejecting the null even under false-discovery-rate control.

²⁰in control-group standard-deviation units

²¹Index construction details and full estimates are reported in Appendix Table 6

4.3.3 Robustness to Individual Thresholds

A key consideration is whether our findings are driven by any single one of the three Census Town criteria. Although each criterion is individually necessary for statutory recognition, eligibility is identified by the *joint* satisfaction of all three thresholds. Figure 1 shows the small individual-threshold jumps relative to the large frontier jump in Figure 2. Further, we estimate the first stage with each threshold *alone*. Individually, these variables yield weak and unstable instruments—the population density and non-agricultural-share criteria have little independent first-stage power in the local sample—whereas the joint instrument is what delivers a strong, stable first stage. Appendix Figure 1 reports the probability of statutory recognition by number of Census Town thresholds met. As suggested by these results, the discontinuity in the probability of statutory recognition is concentrated almost entirely at the point where all three are met, rising from under 4% when two criteria are satisfied to 44% when all three are met.

4.4 Interpretation and Mechanisms

Statutory recognition is best understood as a *bundle*: when a settlement becomes an Urban Local Body, it simultaneously (i) replaces gram-panchayat governance with an elected municipal council and a dedicated urban administration; (ii) gains a new fiscal toolkit—property-tax authority, user charges, access to dedicated urban funding streams and centrally sponsored urban schemes such as JNNURM and AMRUT—while losing access to rural programs such as MGNREGA and PMGSY; and (iii) inherits the Twelfth-Schedule service mandates spanning planning, public health, sanitation, and urban amenities. Our local average treatment effect is the combined effect of this bundle. Relative to the *baseline* of rural governance under a gram panchayat whose institutions and own-source revenues are not designed for urban planning or municipal finance, the bundle represents a discrete upgrade in administrative capacity, fiscal authority, and service obligations.

Although our data does not allow us to separately identify which component of the bundle

does the work, we can offer suggestive evidence on the underlying channels. The consistent pattern in our results across outcomes is informative. Schools, health facilities, and community amenities are services that ULBs are directly mandated to provide, so their expansion is consistent with municipal provision. Financial institutions, by contrast, are *not* a direct municipal function, yet cooperative banks and agricultural credit societies also increase. This is difficult to reconcile with direct provision and points instead toward an indirect channel in which statutory status signals administrative permanence and creditworthiness and raises local demand—making newly recognized towns more attractive locations for formal financial institutions (Burgess and Pande, 2005). The effect of urban governance therefore appears to operate both through direct provision of mandated public goods and through the broader administrative and economic signal triggered by municipalization.

Finally, the effects operate through statutory recognition itself rather than direct Census Town eligibility. Among settlements that remained non-statutory through 2011, the association between CT eligibility and outcomes is uniformly smaller than the full-sample reduced form, with the same pattern across outcomes²².

5 Conclusion

This paper provides evidence that local urban governance status leads to meaningful improvements in local public goods at India’s urbanizing fringe. Using a novel multi-threshold fuzzy RD design that leverages Census Town eligibility criteria, we isolate the effect of local urban governance status on public goods provision from underlying confounding factors.

Our local IV estimates show that statutory recognition increases the number of government schools, hospitals, family welfare centers, cooperative banks, and agricultural credit societies, while decreasing the number of sports facilities, consistent with the idea of realloca-

²²For more details, see Appendix Table 8. Because this comparison conditions on a post-instrument outcome, we treat it as a diagnostic rather than a clean estimate. Nonetheless, it is consistent with statutory municipalization, not eligibility alone, being the operative channel. Fully decomposing the bundle—separating the governance, fiscal, and capacity channels using municipal finance data and longer time horizons—is an important question we leave to future work.

tion of space and land constraints in formalizing settlements. These local average treatment effects indicate that ULB status materially shifts the stock of public infrastructure in settlements near the Census Town thresholds. The corresponding reduced-form estimates are directionally consistent and confirm that crossing the eligibility frontier improves public goods provision even before scaling by the first stage.

The policy implications are twofold. First, there are clear benefits to timely municipalization of emerging urban areas. Our results show that delaying the recognition of urban settlements may imply missed opportunities for public infrastructure. Proactively converting eligible large villages into statutory towns and providing them with fiscal support and a governance framework could help improve their overall infrastructure. Second, our results highlight the importance of state capacity and support for new ULBs. We emphasize that simply declaring a settlement as a municipality is not a panacea. The improvements we document likely come from a combination of federal, state, and local initiatives, the ability to levy local taxes, greater funding availability, and improved local governance. State governments could enhance the benefits of municipalization by ensuring that newly formed ULBs receive adequate resources, technical assistance, and training to fulfill their functions.

Our study opens avenues for future research. One extension would be to examine longer-term outcomes beyond 2011, using household survey data such as subsequent rounds of NFHS to assess impacts on education, health, and income. A clear need remains to understand *why* and *how* local urban governance status leads to improved public goods provision. Is it primarily through increased fiscal resources, improved administrative capacity, greater political representation, or a combination of these factors? We encourage future work to unpack the mechanisms through which municipalization translates into better infrastructure and services. Another avenue is to explore the fiscal channel: how do municipal finances change at the threshold, and to what extent do new ULBs rely on own-source revenue versus transfers? Understanding heterogeneity in statutory recognition across states, such as why some are reluctant to create new towns, possibly to avoid sharing revenue or due

to political patronage networks, could inform policymakers seeking to overcome barriers to timely municipalization. Our findings underscore that formally bringing settlements into the urban administrative fold has measurable benefits, and that ensuring institutional change keeps pace with demographic change will be crucial for sustainable development in rapidly urbanizing economies.

Data Availability Statement

The data and code that replicate the findings of this study are openly available in Harvard Dataverse at <https://doi.org/10.7910/DVN/RR7IPJ>. The archived materials were derived from the [SHRUG v2.1 Population Census files and location keys](#), the [Government of India Open Government Data Census Town Directories](#), and the [National Housing Bank statutory-town lists](#).

Declaration of Competing Interests and Funding

No competing interests are declared. No external funding was received for this research.

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Appendix

Descriptive Outcome Levels

Appendix Table 1 reports unadjusted outcome means by statutory status in the full sample and in the close-to-threshold local sample. These comparisons are purely descriptive and are not part of the identification argument. Their value is diagnostic: they show how large the raw full-sample differences are, and how much those differences compress once we restrict attention to settlements near the Census Town thresholds used in the main analysis.

Appendix Table 1: Descriptive Statistics: Outcome Variables by Statutory Status in the Global and Local Samples

Outcome	Global sample			Local sample		
	Non-ST	ST	Difference	Non-ST	ST	Difference
<i>Education</i>						
Primary schools	1.27	15.50	14.23	1.24	4.96	3.72
Middle schools	0.55	7.64	7.09	0.58	2.65	2.07
Secondary schools	0.19	4.30	4.12	0.25	1.64	1.39
<i>Health</i>						
All hospitals	0.01	1.13	1.12	0.02	0.41	0.39
Family welfare centers	0.06	1.18	1.12	0.08	0.53	0.46
<i>Financial access</i>						
Cooperative banks	0.06	1.90	1.83	0.09	0.97	0.88
Agricultural credit societies	0.16	2.21	2.04	0.17	0.92	0.75
<i>Community infrastructure</i>						
Public libraries	0.13	1.02	0.89	0.11	0.43	0.31
Public reading rooms	0.15	1.03	0.88	0.13	0.45	0.32
Cinema halls	0.06	1.73	1.67	0.05	0.13	0.08
Sports infrastructure	0.42	1.06	0.64	0.46	0.49	0.03

Notes: Entries are unadjusted means of each outcome by statutory town status. “Difference” column is the raw difference in means between statutory and non-statutory settlements. The global sample uses the full regression sample with non-missing covariates and outcome data. The local sample uses the close-to-threshold sample from the main analysis. Outcome-specific sample sizes vary slightly because of missingness and match the corresponding samples used in Appendix Table 5. These differences are descriptive only.

The descriptive comparison reveals a contrast between the two samples. In the full sample, statutory settlements have far higher levels of nearly every public good: for example, they have 15.50 primary schools versus 1.27 in non-statutory settlements, 1.90 cooperative banks versus 0.06, and 1.73 cinema halls versus 0.06. Once we move to the local sample, these gaps narrow substantially. However, note that these raw differences are not adjusted for any covariates or fixed effects, and importantly, contain both compliers and non-compliers in the sample²³. Thus, the non-compliers in the local sample may be attenuating the raw

²³Compliers are settlements whose statutory status would be changed by CT eligibility: they would obtain

differences and we should be careful when interpreting the local raw differences as the effect of statutory recognition.

Balance Checks

To assess the validity of our regression discontinuity design, we examine whether settlements just below and just above the Census Town thresholds are comparable on observable characteristics measured in 2001. Appendix Table 2 presents balance tests for key demographic and economic variables across both the full sample (Panel A) and the close-to-threshold sample (Panel B). The variables include log population, log density, share of non-agricultural male workers, literacy rate, main worker rate, and shares of Scheduled Caste and Scheduled Tribe populations.

As expected in the full sample (Panel A), settlements meeting CT criteria differ substantially from those that do not as they are considerably larger, denser, more economically non-agricultural, and have higher literacy rates. However, when we restrict attention to the close-to-threshold sample (Panel B), these differences narrow substantially. While some gaps remain, particularly in population and density, which are among the threshold criteria themselves, the key observation is that settlements just above and below the thresholds are much more similar than the full sample comparison would suggest. Nevertheless, we control for these remaining differences in our regression models by adding these variables along with appropriate interaction terms and fixed effects.

statutory status by 2011 if CT-eligible in 2001, but would not if ineligible. Non-compliers include never-takers, which would not obtain statutory status even if CT-eligible, and always-takers, which would obtain statutory status even without CT eligibility. Under the usual monotonicity assumption, defiers are ruled out.

Appendix Table 2: Balance Checks: 2001 variables by CT Status

Variable	Non-CT	CT	N
<i>Panel A: Full Sample</i>			
Log Population	6.55 (1.23)	9.84 (0.87)	502,179
Log Density	5.69 (1.45)	7.77 (1.12)	502,179
Non-Ag Male Workers	0.26 (0.24)	0.90 (0.15)	502,179
Literacy Rate	0.47 (0.18)	0.67 (0.12)	502,179
Main Worker Rate	0.32 (0.12)	0.28 (0.08)	502,179
SC Share	0.17 (0.21)	0.13 (0.15)	502,179
ST Share	0.16 (0.31)	0.04 (0.15)	502,179
<i>Panel B: Close-to-Threshold Sample</i>			
Log Population	6.42 (1.18)	8.85 (0.92)	37,151
Log Density	5.60 (1.38)	6.37 (1.25)	37,151
Non-Ag Male Workers	0.71 (0.18)	0.83 (0.14)	37,151
Literacy Rate	0.55 (0.16)	0.63 (0.13)	37,151
Main Worker Rate	0.27 (0.11)	0.32 (0.09)	37,151
SC Share	0.19 (0.22)	0.14 (0.17)	37,151
ST Share	0.12 (0.27)	0.05 (0.18)	37,151

Notes: This table presents mean characteristics in 2001 for settlements that did not meet Census Town (CT) criteria versus those that did. Standard deviations are reported in parentheses below means. Panel A shows the full sample; Panel B restricts to settlements close to the CT thresholds (within specified bandwidth). All variables are measured in 2001, prior to potential statutory recognition by 2011.

Bandwidth Sensitivity

Appendix Table 3 presents bandwidth sensitivity analysis for the effect on government primary schools, the outcome with the largest absolute magnitude. We vary the bandwidth multiplier from $0.5\times$ to $2.0\times$ the baseline thresholds and report both the reduced-form (RF) and instrumental variable (IV) estimates at each bandwidth. At narrow bandwidths ($0.5\times$, $0.75\times$), the first stage is too weak for reliable IV estimation as we lack number of treated observations, and the reduced-form estimates are imprecise. However, starting from the baseline bandwidth ($1.0\times$) and above, the first stage becomes strong enough for valid IV estimation, and the reduced-form estimates stabilize around an increase of approximately 1 additional primary school due to crossing the CT eligibility frontier. The IV estimates at these bandwidths are also positive and statistically significant, though they vary in magnitude due to changes in the strength of the first stage and sample composition. Overall, the results suggest that our main findings are robust to reasonable variations in bandwidth choice, while very narrow bandwidths may lack sufficient power for precise estimation.

Appendix Table 3: Bandwidth Sensitivity: Primary School Estimates

Bandwidth	N	$N(\text{ST}=1)$	FS coef	RF est	RF SE	IV est	IV SE
$0.5\times$	1,216	17	0.032	-0.190	(0.226)	-5.92	(8.38)
$0.75\times$	6,434	57	0.013	0.041	(0.353)	3.11	(25.90)
$1.0\times$ (Baseline)	37,143	127	0.071	0.982***	(0.212)	13.86***	(4.00)
$1.25\times$	51,198	234	0.076	0.899***	(0.145)	11.83***	(2.38)
$1.5\times$	74,214	325	0.085	1.009***	(0.126)	11.84***	(2.06)
$2.0\times$	112,856	535	0.110	1.027***	(0.124)	9.30***	(1.36)

Notes: Each row reports estimates from a different bandwidth specification. Bandwidths are multiples of the baseline thresholds: population $\pm k \times 5,000$, density $\pm k \times 400$, non-agricultural male share $\pm k \times 0.20$. “FS coef” is the first-stage coefficient on CT eligibility. “RF” is the reduced-form effect of CT eligibility on primary schools. “IV” is the 2SLS estimate of statutory recognition on primary schools. All specifications include the full set of controls and district fixed effects with district-clustered standard errors. At narrow bandwidths ($0.5\times$, $0.75\times$), the first stage is too weak for reliable IV estimation. The reduced-form estimate is stable at approximately 1 additional primary school across all bandwidths with sufficient power ($\geq 1.0\times$). *** $p < 0.01$.

Reduced-Form Estimates

Appendix Table 4 reports the reduced-form effect of crossing the Census Town eligibility frontier on local public goods provision in the local sample, estimated from Equation (2). **The dependent variable in each row is the count of the corresponding public good in the settlement in 2011** (e.g., the number of government primary schools, of hospitals, of cooperative banks), and the regressor of interest is the binary Census Town eligibility instrument Z_i . These estimates provide the numerator of the Wald ratio underlying the main IV results in the text. The sign pattern matches the IV estimates throughout: education, health, financial access, and selected community amenities improve, while sports infrastructure declines.

Appendix Table 4: Reduced-Form Effects of Census Town Eligibility on Public Goods Provision

Dependent variable (count in 2011)	Reduced-form coefficient on CT eligibility ($\hat{\gamma}_1$)		
	Estimate	SE	N
<i>Education</i>			
Primary schools	0.982***	(0.212)	37,143
Middle schools	0.572***	(0.121)	37,099
Secondary schools	0.346***	(0.066)	37,148
<i>Health</i>			
All hospitals	0.179***	(0.036)	37,091
Family welfare centers	0.212***	(0.041)	37,077
<i>Financial access</i>			
Cooperative banks	0.297***	(0.044)	37,089
Agricultural credit societies	0.202***	(0.061)	37,093
<i>Community infrastructure</i>			
Public libraries	0.074**	(0.030)	37,147
Public reading rooms	0.102***	(0.035)	37,147
Cinema halls	0.056*	(0.030)	37,146
Sports infrastructure	-0.405***	(0.057)	37,146
Controls: Yes District FE: Yes SEs clustered by district			

Notes: Reduced-form estimates from OLS regressions of each outcome on Census Town eligibility in 2001 (a binary indicator for meeting all three CT thresholds). The local sample restricts to settlements near the 2001 census thresholds (population $\pm 5,000$; density ± 400 ; non-agricultural male share ± 0.20). All specifications include controls for log population, log density, non-agricultural male workforce share, literacy rate, main worker share, and caste composition (SC and ST shares), as well as district fixed effects. Robust standard errors clustered at the district level are in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

In magnitude, the reduced-form estimates are smaller because only a subset of CT-eligible settlements is ultimately statutorily recognized. Even so, the effects remain consistent with IV estimates: crossing the eligibility frontier raises the number of primary schools, middle schools, and secondary schools, while also increasing hospitals, family welfare centers, cooperative banks, and agricultural credit societies. The decline in sports infrastructure is also visible in reduced form, consistent with the land-reallocation interpretation discussed in the main text.

OLS versus IV Estimates

Appendix Table 5 takes the comparison one step further by moving from raw descriptive differences to adjusted regression estimates. Column (1) reports global OLS estimates from the full sample of 502,110 settlements. Column (2) restricts the OLS to the close-to-threshold local sample used in the main analysis. Column (3) reports the local 2SLS estimates from Tables 4–7, where statutory status is instrumented by Census Town eligibility. Reading alongside Appendix Table 1, the progression is straightforward: global raw differences are large, local raw differences are smaller, local OLS attenuates further after conditioning on observables, and the local IV estimates provide the paper’s causal parameter.

Taken together, Appendix Tables 1 and 5 reveal three instructive patterns. First, both the raw means and the global OLS estimates in Column (1) show very large positive differences between statutory and non-statutory settlements for nearly all outcomes. Those differences are informative about broad urban-rural contrasts, but they are heavily contaminated by selection: larger, denser, and more economically developed settlements are both more likely to be statutory towns and more likely to have schools, hospitals, and banks, regardless of their governance status. The global comparison therefore conflates the effect of statutory recognition with pre-existing differences across places.

Second, restricting the comparison to the close-to-threshold local sample sharply compresses the differences. This is visible in the raw means above and again in the adjusted local OLS estimates in Column (2). For education outcomes, the OLS coefficient on primary schools drops from 11.20 to 2.08, and for middle schools from 5.22 to 1.10. For community infrastructure, the local OLS estimates are essentially zero — public libraries (-0.01), reading rooms (-0.08), and cinema halls (-0.01) show no conditional association with statutory status. This attenuation is exactly what we would expect if settlements near the thresholds are much more comparable than the full sample.

Third, Column (3) shows that the local IV estimates that are interpreted as our paper’s causal estimates for settlements whose statutory recognition status was shifted by crossing the Census Town thresholds. These 2SLS coefficients are uniformly larger than the local OLS estimates, and for several outcomes they are larger than even the global OLS estimates. For example, the IV estimate for family welfare centers (3.00) is more than 3.5 times the global OLS (0.85), and the IV estimate for public libraries (1.05) is significant where the local OLS is essentially zero. This pattern is consistent with a LATE interpretation: statutory recognition has especially large effects for marginal settlements near the eligibility frontier, while naive cross-sectional comparisons either mix together incomparable places or understate the effect because of attenuation in the observed status measure.

Appendix Table 5: OLS versus IV Estimates of the Effect of Statutory Recognition

	(1) OLS Global	(2) OLS Local	(3) 2SLS Local
<i>Education</i>			
Primary schools	11.20*** (0.95)	2.08*** (0.49)	13.86*** (4.00)
Middle schools	5.22*** (0.42)	1.10*** (0.22)	7.72*** (2.25)
Secondary schools	3.14*** (0.24)	0.92*** (0.10)	4.89*** (1.30)
<i>Health</i>			
All hospitals	1.00*** (0.06)	0.36*** (0.06)	2.53*** (0.69)
Family welfare centers	0.85*** (0.06)	0.33*** (0.05)	3.00*** (0.88)
<i>Financial access</i>			
Cooperative banks	1.46*** (0.10)	0.64*** (0.07)	4.09*** (0.97)
Agricultural credit societies	1.47*** (0.26)	0.47*** (0.09)	2.84*** (1.01)
<i>Community infrastructure</i>			
Public libraries	0.51*** (0.06)	−0.01 (0.06)	1.05** (0.50)
Public reading rooms	0.49*** (0.06)	−0.08 (0.07)	1.44** (0.63)
Cinema halls	1.49*** (0.11)	−0.01 (0.04)	0.79 (0.49)
Sports infrastructure	−0.09 (0.09)	−0.46*** (0.08)	−5.71*** (1.48)
Controls	Yes	Yes	Yes
District FE	Yes	Yes	Yes
Observations	502,110	37,148	37,148

Notes: Columns (1) and (2) report OLS estimates of the coefficient on statutory town status (ST_i). Column (3) reports 2SLS estimates where ST_i is instrumented by Census Town eligibility in 2001 (Z_i). All specifications include controls for log population, log density, non-agricultural male workforce share, literacy rate, main worker rate, and caste shares, as well as district fixed effects. Robust standard errors clustered at the district level are in parentheses. The “Global” sample includes all settlements with non-missing covariates. The “Local” sample restricts to settlements within $\pm 5,000$ of the population threshold, ± 400 of the density threshold, and ± 0.20 of the non-agricultural share threshold.

Significance: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Multiple Hypothesis Testing

Appendix Table 6 reports [Anderson \(2008\)](#) inverse-covariance-weighted (ICW) indices for the four outcome families. Each index standardizes its components in control-group standard-deviation units, weights them by the inverse of their covariance matrix, and is re-standardized so that estimates are interpretable in control-group SD units. All four families show positive, statistically significant effects in both reduced form (intent-to-treat) and 2SLS.

Appendix Table 6: Anderson (2008) Inverse-Covariance-Weighted Family Indices

Family index (control-group SD units)	Reduced form (ITT)		2SLS (LATE)	
	Estimate	SE	Estimate	SE
Education (primary, middle, secondary)	0.90***	(0.17)	12.20***	(3.29)
Health (hospitals, family welfare centers)	1.29***	(0.23)	18.15***	(4.84)
Financial access (coop. banks, ag. credit)	0.90***	(0.14)	12.77***	(3.16)
Cultural amenities (libraries, reading rooms, cinema)	0.32***	(0.11)	4.57**	(2.01)

Notes: Inverse-covariance-weighted indices following [Anderson \(2008\)](#), in control-group SD units. “Reduced form” regresses the index on CT eligibility. “2SLS” instruments ULB status with CT eligibility. Results from local sample and include controls and district fixed effects. SEs clustered by district. Sports infrastructure is reported separately and is not included in any index. *** $p < 0.01$, ** $p < 0.05$.

Appendix Table 7 reports per-outcome reduced-form effects alongside multiplicity-adjusted p-values: the [Romano and Wolf \(2005\)](#) stepdown familywise-error-rate adjustment computed via a district-level cluster bootstrap so that cross-outcome correlation is respected and [Anderson \(2008\)](#) sharpened two-stage false-discovery-rate q-values. Every main outcome survives the Romano–Wolf adjustment at the 5% level except cinema halls, and all outcomes have sharpened q-values below 0.012.

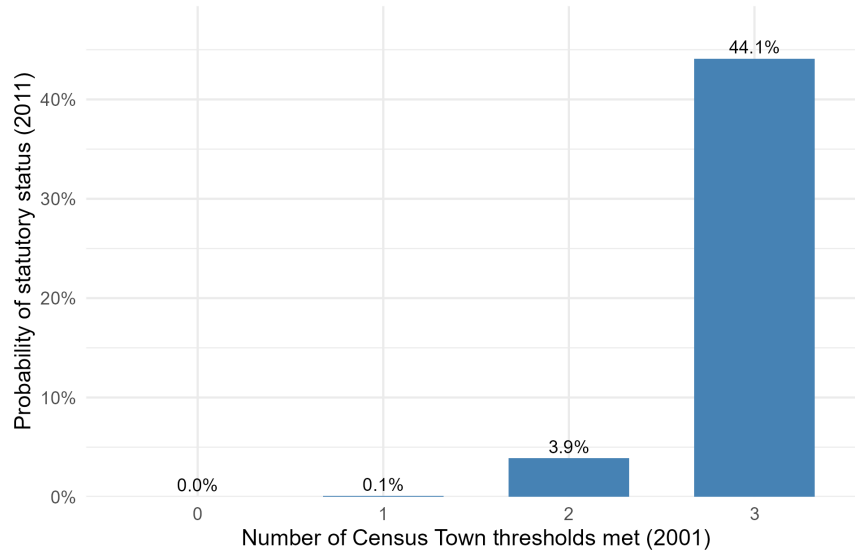
Appendix Table 7: Multiple-Hypothesis-Testing-Adjusted Significance of Reduced-Form Effects

Outcome	RF coef	SE	p (unadj.)	p (Romano–Wolf)	q (sharpened FDR)
Primary schools	0.982	(0.212)	<0.001	0.004	<0.001
Middle schools	0.572	(0.121)	<0.001	0.004	<0.001
Secondary schools	0.346	(0.066)	<0.001	0.001	<0.001
Hospitals	0.179	(0.036)	<0.001	0.004	<0.001
Family welfare centers	0.212	(0.041)	<0.001	0.004	<0.001
Cooperative banks	0.297	(0.044)	<0.001	<0.001	<0.001
Agr. credit societies	0.202	(0.061)	0.001	0.048	<0.001
Public libraries	0.074	(0.030)	0.013	0.048	0.004
Reading rooms	0.102	(0.035)	0.004	0.039	0.002
Cinema halls	0.056	(0.030)	0.061	0.111	0.011
Sports infrastructure	−0.405	(0.057)	<0.001	<0.001	<0.001

Notes: Reduced-form coefficient on CT eligibility. Results from local sample and include controls and district fixed effects. SEs clustered by district. “Romano–Wolf” p -values use the [Romano and Wolf \(2005\)](#) stepdown procedure via a district cluster bootstrap with $N = 999$. “Sharpened FDR” q -values follow the [Anderson \(2008\)](#) two-stage procedure ([Benjamini, Krieger and Yekutieli, 2006](#)).

Sensitivity to Individual Thresholds

This section documents that our findings are not driven by any single Census Town criterion. Census Town eligibility is, by construction, the *joint* satisfaction of all three thresholds, and the discontinuity in the probability of statutory recognition is concentrated at the point where all three are met. Appendix Figure 1 plots the probability of statutory status by the number of thresholds a settlement satisfies in 2001: the probability is essentially zero for settlements meeting zero or one criterion (0.01% and 0.08%), rises modestly to 3.9% at two criteria, and then jumps to 44.1% when all three are met. The action is overwhelmingly at the joint frontier.



Appendix Figure 1: Probability of statutory status by number of Census Town thresholds met (2001)

Exclusion Restriction: Never-Municipalized Settlements

As a stress test of the exclusion restriction and a probe of mechanism, Appendix Table 8 compares the main reduced-form effect of CT eligibility, the association between CT eligibility and outcomes among settlements that *never* became statutory, and the main 2SLS estimate. If CT eligibility affected outcomes only through channels other than statutory recognition, the never-municipalized effects would resemble the full-sample reduced form. Instead they are uniformly smaller (e.g., 0.81 vs. 0.98 primary schools), consistent with statutory recognition—not eligibility per se—being the operative channel. Because the never-municipalized sample conditions on a post-instrument outcome, we treat this as a diagnostic rather than a clean causal estimate.

Appendix Table 8: Reduced Form vs. Never-municipalized vs. 2SLS estimates

Outcome	Main reduced form	Never-municipalized	Main 2SLS
Primary schools	0.982*** (0.212)	0.810*** (0.126)	13.86*** (4.00)
Middle schools	0.572*** (0.121)	0.511*** (0.099)	7.72*** (2.25)
Secondary schools	0.346*** (0.066)	0.304*** (0.063)	4.89*** (1.30)
Hospitals	0.179*** (0.036)	0.159*** (0.037)	2.53*** (0.69)
Family welfare centers	0.212*** (0.041)	0.211*** (0.044)	3.00*** (0.88)
Cooperative banks	0.297*** (0.044)	0.264*** (0.045)	4.09*** (0.97)
Agr. credit societies	0.202*** (0.061)	0.173*** (0.062)	2.84*** (1.01)
Public libraries	0.074** (0.030)	0.095*** (0.032)	1.05** (0.50)
Reading rooms	0.102*** (0.035)	0.130*** (0.037)	1.44** (0.63)
Cinema halls	0.056* (0.030)	0.071** (0.032)	0.79 (0.49)
Sports infrastructure	-0.405*** (0.057)	-0.381*** (0.060)	-5.71*** (1.48)

Notes: “Main reduced form” regresses each outcome on CT eligibility in the full local sample. “Never-municipalized” restricts to settlements that remained non-statutory through 2011. “Main 2SLS” reproduces the local IV estimates from the main text. Controls, district FE, district-clustered SEs throughout. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.